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Concept: NCERT Number Systems Q1 — source-linked TEST Core2/Core1A

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NCERT Number Systems Q1 — source-linked TEST Core2/Core1A: Construction

Contents and concept route



Every rational number is real

Subject: TEST

Topic: Number Systems: rational numbers and real numbers

Concept: Every rational number is real

Concept difficulty: Easy concept

1

Distinguish a universal containing set from narrower alternatives using a rational counterexample.

Easy concept

Section 1 of 1

Conditions for this construction

- Distinguish integers from fractions with nonzero denominators.
- Recognise negative rational numbers.

Why this section matters for the question set

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1 Define rational numbers as quotients a/b of integers with b nonzero.

Why valid: Every such quotient is a real number on the number line.

Result: Any rational belongs to the real numbers.

2 Test the narrower claims with $-1/2$, $1/2$ and -2 .

Why valid: These are rational, but not all are whole, natural or integer values.

Result: Reject natural number, integer and whole number as universal classifications.

3 The step the hard question turns on **Identify the real-number set as a valid universal superset of the rational numbers.**

Why valid: The set inclusion $Q \subseteq R$ establishes option C without claiming $R \subseteq Q$.

Result: Every rational number is a real number.

The key step

Use $Q \subseteq R$ as an inclusion relation, not the false reverse inclusion $R \subseteq Q$.

Formula reference · meaning and conditions

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Worked explanation

Unit 1 Ex 1.1 Q1 · Verified curricular source, faithful NCERT wording

Every rational number is

Follow the operation, its justification and its result.

- 1. Interpret 'Every rational number is' as asking for a set containing all rational numbers.**

Why valid: A universal claim must hold for negative and fractional rationals, not just positive whole examples.

Result: Find a valid universal superset of $\mathbb{Q}$.

2. **Use $\mathbb{Q} \subseteq \mathbb{R}$: every rational a/b with integer a , $b \neq 0$ represents a real number.**

Why valid: Every quotient of integers with nonzero denominator is a real number.

Result: Option (C) is valid.

3. **Test 1/2 against natural, integer and whole-number categories.**

Why valid: One nonintegral rational refutes each narrower universal option.

Result: Reject the narrower categories; retain (C).

Result

(C) a real number. Every rational number is a real number.

Check

Independent definition and counterexample check: $1/2$ is rational and real, but not a natural number, integer or whole number; $\sqrt{2}$ is real but irrational, so reverse inclusion does not follow.

Mistake clinic · diagnose and repair



Quick check

1 · CHECK

Use $1/2$: it is rational and real, but not integer, whole or natural. Therefore three narrower universal choices fail.

2 · CHECK

Use $\sqrt{2}$ to disprove the converse: a real number need not be rational.

Now you do one

Is $-3/2$ rational and real? Is every real number rational? Justify both claims.

Practice this concept in Core2

Unit 1 Ex 1.1 Q1

Next steps

Practice this concept →

All questions in Question Bank →

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